

TODO: Title of the multimodal cancer prediction study

TODO: author names
TODO: institutions and addresses
TODO: corresponding author and email

Abstract

Background: [TODO: State the cancer prediction task, motivation, study objective, and model name.]
Results: [TODO: Summarize the completed comparisons in Tables 1 and 2, if Table 2 is retained; insert only documented results.]
Conclusions: [TODO: State conclusions supported by the completed comparisons. Do not insert a performance or novelty claim yet.]

Keywords: Multimodal learning; cancer prediction; textual data; genomic data; multimodal fusion

1 Background

[TODO: Introduce the clinical prediction task and the motivation for combining textual and genomic information.]

The Cancer Genome Atlas (TCGA) provides the data-source context for this draft [?]. [TODO: Describe the specific data used and the research question.]

The baseline reporting framework uses Table 1 of Song et al. [?] as the planned source of published comparison values. [TODO: Add the study rationale and a factual description of Ours once the method has been specified.]

2 Methods

2.1 Data and input checklist

- Cohort columns: BLCA, BRCA, GBMLGG, LUAD, UCEC, CRAD, HNSC, and STAD. [TODO: Record cohort definitions, sample counts, inclusion criteria, and data versions.]
- Two-modality inputs: text and genomics. [TODO: Record data fields, preprocessing, and feature representations.]
- Optional three-modality inputs: text, genomics, and [TODO: third modality and its representation].
- Prediction target: [TODO: survival or classification, outcome definition, labels, and prediction time point if applicable].

2.2 Baseline comparison checklist

- Table 1: Cross-Attn Fusion, Early Fusion, Late Fusion, and Ours.
- Table 2, if retained: LIMOE, MAGGate, MulT, TF, Cross-Attn Fusion, Early Fusion, Late Fusion, and Ours.

- Ours: [TODO: model name, existing implementation identifier, and a short factual method description].

Bibliographic slots are reserved for Cross-Attn Fusion, Early Fusion, and Late Fusion [? ? ?], and for LIMOE, MAGGate, and Transformer Fusion (TF) [? ? ?]. MulT has a candidate original reference [?]. [TODO: Resolve each TODO-CITE and confirm the mapping from each table label to its cited method.]

2.3 Evaluation and filling checklist

- Validation for Ours: five-fold evaluation. [TODO: Record split files, fold identifiers, and their relationship to the source study.]
- Per-cohort reporting: fill the eight cohort columns in Table 1 and, if retained, Table 2.
- Metric: [TODO: full metric name, abbreviation, implementation, and whether higher or lower values are better].
- Summary within each cohort: [TODO: aggregation across folds and the uncertainty measure reported beside the point estimate].
- Overall: [TODO: aggregation definition, weighting, and source or calculation record for each row].
- Ours result records: [TODO: paths to existing per-fold results, configurations, and the checkpoint-selection rule].

2.4 Provenance of comparison values

Non-Ours values in Tables 1 and 2 are planned to be transcribed from Table 1 of Song et al. [?]. These are literature-reported results; they are not experiments reproduced by the authors of this study. Ours rows are reserved for the authors’ own experimental results.

[TODO: Record the source row, column, and setting for each transcribed value. Record the calculation used for any derived Overall entry.] All numerical entries remain unfilled in this draft.

3 Results

3.1 Two-modality comparison

[TODO: Fill Table 1 from the designated sources and the existing Ours results, then describe the observed per-cohort and Overall comparisons.]

Table 1: Two-modality comparison using text and genomics. Metric and uncertainty summary: TODO.

Method	BLCA	BRCA	GBMLGG	LUAD	UCEC	CRAD	HNSC	STAD	Overall
Cross-Attn Fusion	–	–	–	–	–	–	–	–	–
Early Fusion	–	–	–	–	–	–	–	–	–
Late Fusion	–	–	–	–	–	–	–	–	–
Ours	–	–	–	–	–	–	–	–	–

Non-Ours values are planned to be taken from Table 1 of Song et al. [?], not reproduced in this study. Ours values will be supplied from the authors’ own experiments. The symbol – denotes an unfilled entry. Overall: TODO, including whether it is transcribed or calculated.

3.2 Three-modality comparison: optional

[TODO: Confirm whether Table 2 is retained and specify the third modality. If retained, fill its entries and describe the comparisons.]

Table 2: Optional three-modality comparison using text, genomics, and a third modality (TODO). Metric and uncertainty summary: TODO.

Method	BLCA	BRCA	GBMLGG	LUAD	UCEC	CRAD	HNSC	STAD	Overall
LIMOE	–	–	–	–	–	–	–	–	–
MAGGate	–	–	–	–	–	–	–	–	–
MulT	–	–	–	–	–	–	–	–	–
TF	–	–	–	–	–	–	–	–	–
Cross-Attn Fusion	–	–	–	–	–	–	–	–	–
Early Fusion	–	–	–	–	–	–	–	–	–
Late Fusion	–	–	–	–	–	–	–	–	–
Ours	–	–	–	–	–	–	–	–	–

Non-Ours values are planned to be taken from Table 1 of Song et al. [?], not reproduced in this study. Ours values will be supplied from the authors’ own experiments. TF denotes Transformer Fusion. The symbol – denotes an unfilled entry. Overall: TODO, including whether it is transcribed or calculated.

4 Discussion

[TODO: Interpret the completed comparisons without inventing a mechanistic explanation or claiming an unmeasured improvement.]

[TODO: Discuss the findings in relation to the cited methods, distinguishing literature-reported values from the authors’ results.]

[TODO: State the limitations supported by the final study design and completed evidence.]

5 Conclusions

[TODO: Summarize the findings and their scope after the result tables have been completed.]

Abbreviations

TCGA The Cancer Genome Atlas.

TF Transformer Fusion.

Cross-Attn Cross-Attention.

[TODO: Add the full names of the reported metric, cohort labels, and remaining method abbreviations; confirm the definition of CRAD.]

Declarations

Ethics approval and consent to participate

[TODO: Author-confirmed ethics and consent statement, including approval, exemption, or applicability details.]

Consent for publication

[TODO: Author-confirmed consent-for-publication statement.]

Availability of data and materials

[TODO: Dataset access details and links to available code, split files, and result records. State any actual access restrictions.]

Competing interests

[TODO: Author-confirmed declaration of competing interests.]

Funding

[TODO: Funding bodies, grant identifiers, and funder roles, or an author-confirmed statement of no funding.]

Authors' contributions

[TODO: Author names or initials and their actual contributions.]

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